

Lab8: Insight into MVC and Laminas

Objective

- Install and configure Composer on macOS or Windows
- Set up a basic Laminas project by following the official tutorial
- Understand and explain the core principles of the MVC architecture

Tips

- These labs are expected to take more than the allocated **4 hours (Week 12 – Week 13)**. You should try the tasks before the lab. Practice makes perfect.
- Before the lab, you could post your questions in LMO forum, and you are welcome to offer solutions to the questions in LMO.
- During the lab, you could discuss with TAs for your questions. Face to face discussion will have a relatively high efficiency.

A. Brief Introduction

1. **Composer** is a dependency manager for PHP, simplifying the process of managing libraries and packages used in PHP projects. It allows developers to easily install, update, and manage various code components their projects rely on.
2. **Laminas**, on the other hand, is a powerful PHP framework. It provides a structured way to build web applications, following modern development practices. Laminas offers a wide range of components/tools that help in tasks such as routing requests, handling user input, interacting with databases, and rendering views.

B. Install Composer

1. This is for **macOS**:
 - (1) **Open Terminal**: You can locate the **Terminal** application in the "Utilities" folder within the "Applications" folder
 - (2) **Install Homebrew**:

- First, check if you have already installed Homebrew by running:
`brew --version`
- If version is shown, please jump to (3)
- Install Homebrew by running the following command (copy all three lines together):
`/bin/bash -c "$(curl -fsSL
https://raw.githubusercontent.com/Homebrew/install/HEAD/
install.sh)"`
- Now, check if Homebrew is properly installed:
`brew --version`

(3) Install PHP with version 8.2:

- Install PHP using Homebrew:
`brew install php@8.2`
- Add PHP to your PATH with three commands:
 - `echo 'export
PATH="/usr/local/opt/php@8.2/bin:$PATH"' >> ~/.zshrc`
 - `echo 'export
PATH="/usr/local/opt/php@8.2/sbin:$PATH"' >> ~/.zshrc`
 - `source ~/.zshrc`
- Now, you could check the PHP version by running:
`php -v`

(4) Install Composer:

- Download the Composer installer using the following command:
`curl -sS -k https://getcomposer.org/installer | php`
- Move the "composer.phar" file to a directory in your path so that you can use the `composer` command globally:
`sudo mv composer.phar /usr/local/bin/composer`

(5) Verify Installation:

- To check if Composer is installed correctly, run the following command:
`composer --version`

2. This is for Windows:

(1) **Open Command Prompt as Administrator:** Right-click on the **Command Prompt** icon and select "Run as administrator"

(2) **Set the Path to XAMPP PHP:**

- If you installed XAMPP in the default location (**C:\xampp**), the PHP executable is at **C:\xampp\php**
- Add this path to the system's **PATH** environment variable:
 - Go to **Control Panel > System and Security > System**
 - Click on **Advanced system settings**
 - In **System Properties**, click on **Environment Variables**
 - In **System variables**, find the **Path** variable, select it, and click **Edit**
 - Click **New** and add **C:\xampp\php**
 - Click **OK** to save the changes
- Now, you could check the PHP version in the Command Prompt by running:
`php -v`

(3) **Install Composer:**

- Visit the official Composer website at <https://getcomposer.org/Composer-Setup.exe> and download the installer
- Double-click on the downloaded installer file (**Composer-Setup.exe**), and follow the installation wizard prompts
- When prompted, the installer should automatically detect XAMPP PHP executable (**C:\xampp\php\php.exe**). If not, manually select it

(4) **Verify Installation:**

- Open the command Prompt, and run the following command:
`composer --version`

(5) **Change Configuration:**

- Open **C:\xampp\php\php.ini** with Sublime Text or VS Code
- Remove **;** before the following extensions:
`extension=intl`
`extension=sqlite3`
`extension=zip`

C. Laminas Tutorial

1. Open the terminal:

- On macOS, use **Terminal**
- On Windows, use **Command Prompt**

2. Navigate to the project directory:

- On macOS, run:
`cd /Applications/XAMPP/htdocs/can302`
- On Windows, run:
`cd C:\xampp\htdocs\can302`

3. Follow the Laminas tutorial step by step:

- Open the official guide: <https://docs.laminas.dev/tutorials/getting-started/overview/>
- Progress through the tutorial over the next few weeks: **Overview > The Skeleton Application > Modules > Routing and Controllers > Database and Models > Forms and Actions > Conclusion**
- Additional references:
 - Lecture 7 under "Lecture notes" in LMO
 - <https://docs.laminas.dev/laminas-mvc/intro/>
 - <https://docs.laminas.dev/laminas-mvc/quick-start/>